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Experiment 4: Demonstration of Network Communication Models (OSI and TCP/IP) Using a Network Simulation Tool

1. Aim

To demonstrate how data travels through the layers of the OSI and TCP/IP reference models using a network simulation tool such as Cisco Packet Tracer.

2. Theory

Network communication is organised into layers so that each layer provides a service to the layer above it and uses the service of the layer below it. The OSI model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation and Application. The TCP/IP model used in practice has four layers: Network Access, Internet, Transport and Application. The TCP/IP layers correspond to groups of OSI layers, which makes the OSI model useful for explaining the functions of each protocol.

When an application sends data, each layer adds its own header to the data received from the layer above. This process is called encapsulation. The data unit is called a segment at the transport layer, a packet at the network layer and a frame at the data link layer. At the receiver, each layer removes its header and passes the payload upward, which is called decapsulation.

3. Layer Comparison

OSI Layer	TCP/IP Layer	Function	Example Protocol / Device
7 Application	Application	Network services to user applications	HTTP, DNS, SMTP
6 Presentation	Application	Encoding, encryption, compression	TLS, JPEG
5 Session	Application	Session establishment and management	NetBIOS, RPC
4 Transport	Transport	End-to-end delivery and reliability	TCP, UDP
3 Network	Internet	Logical addressing and routing	IP, ICMP, Router
2 Data Link	Network Access	Frame delivery on a local link	Ethernet, Switch
1 Physical	Network Access	Bit transmission over media	Cables, Hub

4. Simulation Setup

- A small network was built in the simulator with two PCs (PC0 and PC1), a Layer 2 switch and a router connecting to a second LAN with a web server.

- Each PC was given a static IP address, subnet mask and default gateway pointing to the router interface.
- The router interfaces were configured with IP addresses on their respective subnets, and connectivity was confirmed.

5. Procedure

- A ping was issued from PC0 to PC1 on the same LAN and the simulation mode was used to step through each packet.
- A second ping was issued from PC0 to the web server on the other LAN, so the packet crossed the router.
- An HTTP request was generated from a PC browser to the web server, and the PDU (protocol data unit) details were inspected at every layer.
- Each PDU inspector view was noted, including MAC addresses, IP addresses, port numbers and protocol type fields.

6. Observations

Test	Layer Observed	Key Information Seen
Local ping	Data link	Source and destination MAC addresses change only on the local segment
Routed ping	Network	Source and destination IP addresses stay the same across the router
HTTP request	Transport	TCP header with destination port 80 and a sequence number
HTTP request	Application	GET request line and Host header in the payload

The observations confirm that the ICMP echo request is carried in an IP packet, which is carried in an Ethernet frame. The IP header keeps the same source and destination addresses across the router, while the MAC addresses are rewritten at each hop. The TCP header identifies the web service by port 80, showing how the transport layer maps application services to ports.

7. Analysis

The simulation shows that routers operate on the network layer, using IP addresses to choose the next hop, while switches operate on the data link layer, using MAC addresses to forward frames within a LAN. This explains why IP addresses are constant end to end, whereas MAC addresses change at every router interface. The HTTP request demonstrates that an application protocol relies entirely on the lower layers for delivery, reliability and addressing.

8. Limitations

A simulator provides a controlled environment with predictable timing and no real traffic. Physical-layer behaviour such as signal loss, interference and cable faults is not represented. Real

networks also include wireless links, NAT, firewalls and security protocols that are simplified or absent in the model used here.

9. Result and Conclusion

The OSI and TCP/IP layers were mapped to real network activity in the simulator, and encapsulation across layers was observed directly in the PDU inspector. The exercise shows that layered design allows each protocol to be developed, replaced and troubleshot independently, which is the main strength of the layered approach.